HDOC - DAY 11 REPORT

Name: Dhruv Pandey
SAP ID: 590037660

Menu-Driven Digit Operations Using C

Problem Statement

Prepare the algorithm and test case table, then write, compile, execute and test a menu-driven C program using switch-case. The program performs: sum of squares of digits using a while loop, sum of cubes of digits using a for loop, and the difference between the sum of cubes and the sum of squares using a do-while loop.

Algorithm

1. Start the program and display a menu of digit operations.
2. Read the user's menu choice.
3. For choice 1, input a positive integer and use a while loop to find the sum of squares of its digits.
4. For choice 2, input a positive integer and use a for loop to find the sum of cubes of its digits.
5. For choice 3, input a positive integer and use a do-while loop to calculate both sums and print their difference.
6. For choice 4, exit the program; otherwise display an invalid-choice message.
7. Repeat the menu until Exit is selected, then stop.

Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	Choice 1, Number 123	Sum of squares = 14	Sum of squares = 14	Pass
2	Choice 2, Number 123	Sum of cubes = 36	Sum of cubes = 36	Pass
3	Choice 3, Number 123	Difference = 22	Difference = 22	Pass

C Program - Part 1

```
UW PICO 5.09                    File: day111.c                    Modified

#include <stdio.h>

int main() {
    int choice;
    int number, temp, remainder, sum, diff;
    int sum_squares, sum_cubes;

    do {
        printf("\n=== Menu-Driven Digit Operations ===\n");
        printf("1. Find the sum of squares of its digits\n");
        printf("2. Find the sum of cubes of its digits\n");
        printf("3. Find the difference between sum of cubes and sum of squares of its digits\n");
        printf("4. Exit\n");
        printf("Enter your choice (1-4): ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter a positive integer: ");
                scanf("%d", &number);
                if (number < 0) {
                    printf("Error: Please enter a positive integer.\n");
                    break;
                }
                sum = 0;
                temp = number;
                while (temp > 0) {
                    remainder = temp % 10;
                    sum += (remainder * remainder);
                    temp = temp / 10;
                }
                printf("For Number: %d, Sum of squares of digits = %d\n", number, sum);
                break;

            case 2:
                printf("Enter a positive integer: ");
                scanf("%d", &number);
                if (number < 0) {
                    printf("Error: Please enter a positive integer.\n");
                    break;
                }
                sum = 0;
                for (temp = number; temp > 0; temp /= 10) {
                    remainder = temp % 10;
                    sum += (remainder * remainder * remainder);
                }
                printf("For Number: %d, Sum of cubes of digits = %d\n", number, sum);
                break;

            case 3:
                printf("Enter a positive integer: ");
                scanf("%d", &number);
                if (number < 0) {
                    printf("Error: Please enter a positive integer.\n");
                    break;

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

C Program - Part 2

```
  UW PICO 5.09                              File: day111.c                              Modified

                sum += (remainder * remainder);
                temp = temp / 10;
            }
            printf("For Number: %d, Sum of squares of digits = %d\n", number, sum);
            break;

        case 2:
            printf("Enter a positive integer: ");
            scanf("%d", &number);
            if (number < 0) {
                printf("Error: Please enter a positive integer.\n");
                break;
            }
            sum = 0;
            for (temp = number; temp > 0; temp /= 10) {
                remainder = temp % 10;
                sum += (remainder * remainder * remainder);
            }
            printf("For Number: %d, Sum of cubes of digits = %d\n", number, sum);
            break;

        case 3:
            printf("Enter a positive integer: ");
            scanf("%d", &number);
            if (number < 0) {
                printf("Error: Please enter a positive integer.\n");
                break;
            }
            sum_squares = 0;
            sum_cubes = 0;
            temp = number;
            do {
                remainder = temp % 10;
                sum_squares += (remainder * remainder);
                sum_cubes += (remainder * remainder * remainder);
                temp = temp / 10;
            } while (temp > 0);
            diff = sum_cubes - sum_squares;
            printf("For Number: %d, Difference = %d\n", number, diff);
            break;

        case 4:
            printf("Exiting the program. Goodbye!\n");
            break;

        default:
            printf("Invalid choice! Please select an option between 1 and 4.\n");
        }

    } while (choice != 4);

    return 0;
}

^G Get Help    ^O WriteOut    ^R Read File    ^Y Prev Pg    ^K Cut Text      ^C Cur Pos
^X Exit        ^J Justify     ^W Where is     ^V Next Pg    ^U UnCut Text    ^T To Spell
```

Compilation and Execution

```
Last login: Sat Sep  5 22:28:32 on ttys000
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day111.c
dhruvpandey@DHRUVs-MacBook-Air ~ %
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day111.c -o day111
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day111

=== Menu-Driven Digit Operations ===
1. Find the sum of squares of its digits
2. Find the sum of cubes of its digits
3. Find the difference between sum of cubes and sum of squares of its digits
4. Exit
Enter your choice (1-4): 1
Enter a positive integer: 2
For Number: 2, Sum of squares of digits = 4

=== Menu-Driven Digit Operations ===
1. Find the sum of squares of its digits
2. Find the sum of cubes of its digits
3. Find the difference between sum of cubes and sum of squares of its digits
4. Exit
Enter your choice (1-4): 3
Enter a positive integer: 1
For Number: 1, Difference = 0

=== Menu-Driven Digit Operations ===
1. Find the sum of squares of its digits
2. Find the sum of cubes of its digits
3. Find the difference between sum of cubes and sum of squares of its digits
4. Exit
Enter your choice (1-4): 3
Enter a positive integer: 1
For Number: 1, Difference = 0

=== Menu-Driven Digit Operations ===
1. Find the sum of squares of its digits
2. Find the sum of cubes of its digits
3. Find the difference between sum of cubes and sum of squares of its digits
4. Exit
Enter your choice (1-4): 3
Enter a positive integer: 1
For Number: 1, Difference = 0

=== Menu-Driven Digit Operations ===
1. Find the sum of squares of its digits
2. Find the sum of cubes of its digits
3. Find the difference between sum of cubes and sum of squares of its digits
4. Exit
Enter your choice (1-4):
```